

PARUL UNIVERSITY

Faculty of Engineering & Technology

B.Tech – Semester III

Subject: Design of Data Structures (303105201)

Time: 2.5 Hours

Maximum Marks: 60

SECTION – A

(Total: 30 Marks)

Q1 (Compulsory) – 12 Marks

a) Answer ANY THREE of the following: ($3 \times 2 = 6$)

1. Define Time Complexity.
2. What is Stack Underflow?
3. Define Recursion.
4. What is a Priority Queue?

b) Answer ANY THREE of the following: ($3 \times 2 = 6$)

1. Define Binary Search Tree.
 2. What is Balance Factor?
 3. Define Hash Function.
 4. What is Load Factor?
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Q2 – 9 Marks

a) Explain recursion using Tower of Hanoi problem. (4)

b) Answer ANY ONE of the following: (5)

- Write recursive algorithm for Factorial and explain it.
 - Write recursive algorithm for Fibonacci series.
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Q3 – 9 Marks

- a) Explain Quick Sort algorithm. (4)
- b) Answer ANY ONE of the following: (5)
- Sort the following array using Quick Sort and show all steps:
45, 12, 78, 34, 23
 - Explain best, worst and average case time complexity of Quick Sort.
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SECTION – B

(Total: 30 Marks)

Q4 (Compulsory) – 12 Marks

- a) Answer ANY THREE of the following: ($3 \times 2 = 6$)
1. Define Linked List.
 2. What is a Circular Queue?
 3. Define AVL Tree.
 4. What is Breadth First Search (BFS)?
- b) Answer ANY THREE of the following: ($3 \times 2 = 6$)
1. Define Binary Tree.
 2. What is Collision in hashing?
 3. Define Depth First Search (DFS).
 4. What is an Adjacency List?
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Q5 – 9 Marks

- a) Explain insertion and deletion operations on singly linked list. (4)
- b) Answer ANY ONE of the following: (5)
- Explain Doubly Linked List with suitable diagram.
 - Explain Header Linked List and mention its advantages.

Q6 – 9 Marks

- a) Explain AVL Tree rotations. (4)
- b) Answer ANY ONE of the following: (5)
- Insert the following values into an AVL Tree:
30, 20, 40, 10, 25 and show rotations.
 - Explain applications of AVL Tree.